

RRC-Raptor-V4H

- Switch

V0.0.3

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i. VERSION LIST

Description	Version	Date	Author
Switch	V0.0.1	2024/08/30	KevinLin
Modify 4.1.1(Page11) 、 4.1.4(Page14) 、 4.4.2(Page20) 、 4.4.3(Page21) 、 4.4.6(Page23) 、 4.6.2(Page27) 、	V0.0.2	2024/10/30	KevinLin
File attribute changes	V0.0.3	2025/01/22	kevinlin

ii. HASHTAG

#V4H
#Switch
#DipSwitch
#Button
#Variable Voltage

1. Overview

Target :

DIP switches, buttons and variable voltage distribution, introduction and initial settings in Raptor-V4H.

2. Configuration Diagram

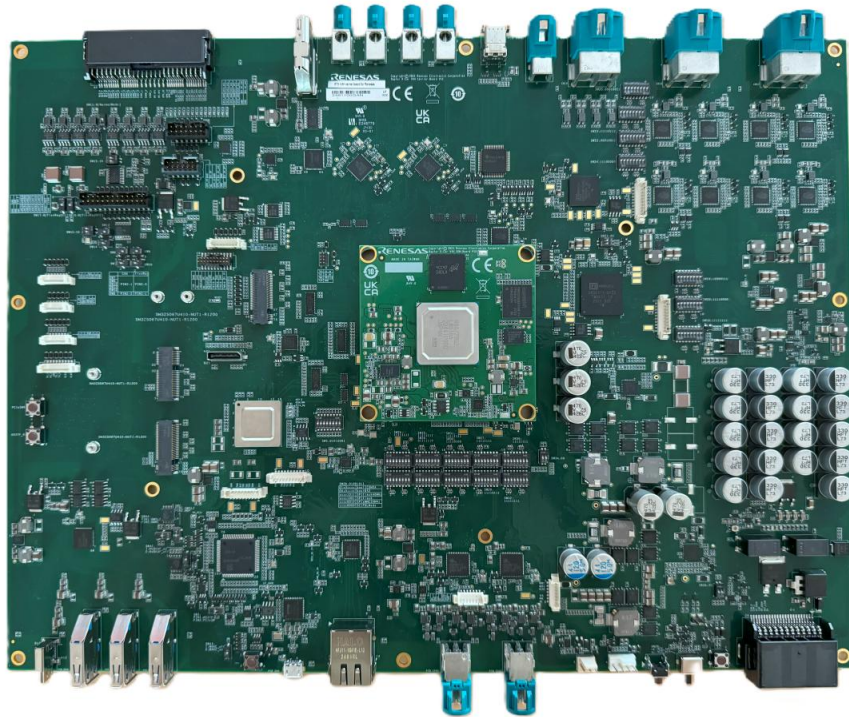


Figure 1. Top View

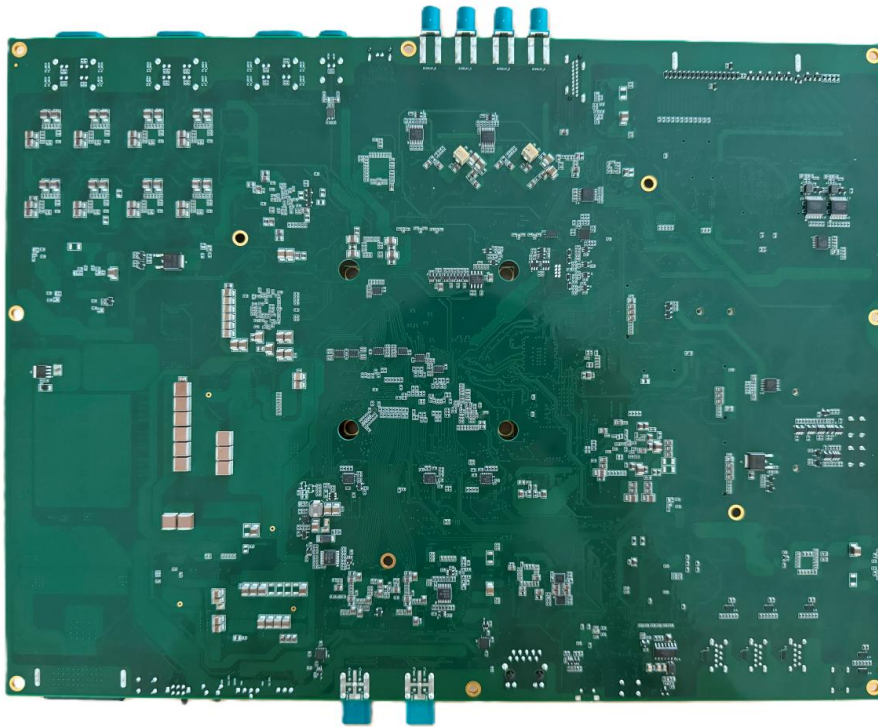


Figure 2. Bottom View

3. DIP SWITCH

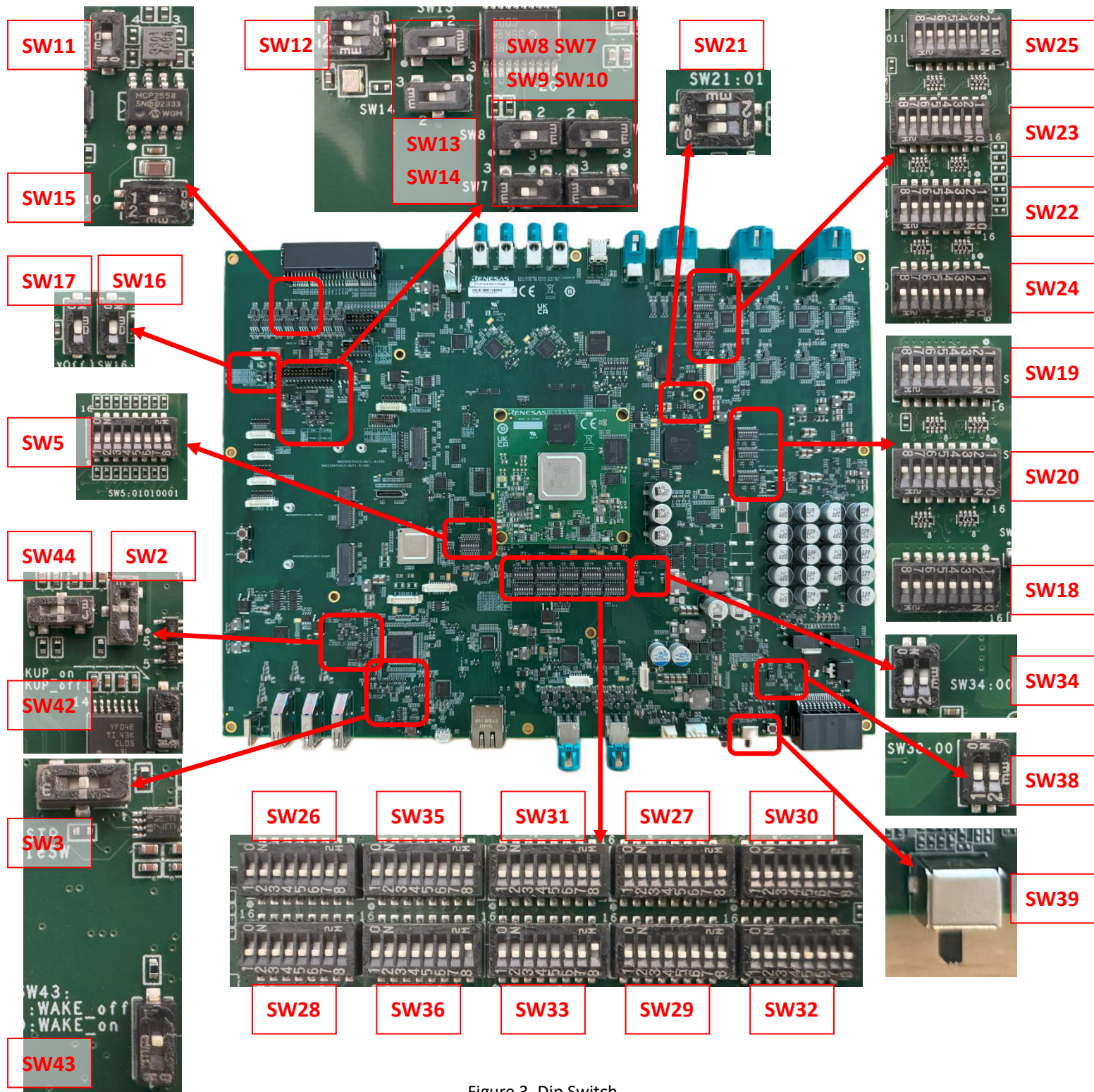


Figure 3. Dip Switch

4. SWITCH

Default Setting :

◆ Boot

SW26	SW35	SW31	SW27	SW30
1111 1111	1111 1111	1111 1111	1111 1111	1111 1111
SW28	SW36	SW33	SW29	SW32
0100 1011	0111 1110	1111 0100	1111 0111	1111 1111

◆ PCIe

SW5	SW3
01000001	center

◆ SPI

SW34	SW38
00	00

◆ Ethernet

SW18	SW19	SW20	SW21
1111 1111	1000 0110	0000 0000	01
SW22	SW23	SW24	SW25
0001 0011	0000 0110	1110 0000	0001 0001

◆ Display

SW2	SW44
center	center

◆ CANFD & FlexRay

SW7	SW8	SW9	SW10	SW13	SW14
PIN 2 – 1	PIN 2 – 3	PIN 2 – 1	PIN 2 – 1	PIN 2 – 1	PIN 2 – 1
SW11	SW12	SW15	SW16	SW17	
1	10	10	0	0	

◆ MCU & Power

SW39	SW42	SW43
0	0	0

4.1. BOOT MODE SWITCH

4.1.1. SW26 & SW28

SW26	1:MD7	2:MD6	3:X	4:MD8	5:MD4	6:MD3	7:MD2	8:MD1
DEFAULT	ON	ON	ON	ON	ON	ON	ON	ON
ON	PullHigh	PullHigh	X	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh
OFF	Open	Open	X	Open	Open	Open	Open	Open
SW28	1:MD7	2:MD6	3:X	4:MD8	5:MD4	6:MD3	7:MD2	8:MD1
DEFAULT	OFF	ON	OFF	OFF	ON	OFF	ON	ON
ON	GND	GND	X	GND	GND	GND	GND	GND
OFF	Open	Open	X	Open	Open	Open	Open	Open

◆ [SW28]

1:MD7	2:MD6	Selection of Master Boot Processor
OFF	ON	Booted through ICUMAX
OFF	OFF	Booted through Cortex-R52
Other than above		Setting prohibited

4:MD8	Selection mode
ON	Setting prohibited
OFF	Use this setting

5:MD4	6:MD3	7:MD2	8:MD1	Selection of Boot Device
ON	OFF	ON	ON	Serial Flash boot at single read 40MHz using DMA
ON	OFF	OFF	ON	Serial Flash boot at DDR read 80MHz using DMA
OFF	OFF	ON	OFF	eMMC boot at 50MHz x8 bus widths using DMA
OFF	OFF	OFF	OFF	SCIF downloading mode
Other than above				Setting prohibited

4.1.2. SW35 & SW36

SW35	1:MD5	2:MD20	3:MD11	4:MD10	5:MD35	6:MDT0	7:MD21	8:X
DEFAULT	ON	ON	ON	ON	ON	ON	ON	ON
ON	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X
OFF	Open	Open	Open	Open	Open	Open	Open	X
SW36	1:MD5	2:MD35	3:MDT0	4:MD21	5:MD20	6:MD3	7:MD2	8:X
DEFAULT	OFF	ON	ON	ON	OFF	ON	OFF	OFF
ON	GND	GND	GND	GND	GND	GND	GND	X
OFF	Open	Open	Open	Open	Open	Open	Open	X

◆ [SW36]

1:MD5	2:MD35	Selection mode
ON	-	Setting prohibited
OFF	-	Normal boot
-	ON	Use this setting
-	OFF	Setting prohibited

3:MDT0	4:MD21	5:MD20	6:MD11	7:MD10	Main_JTAG	
ON	ON	ON	ON	ON	-	Normal
ON	ON	ON	OFF	ON	ICUMX JTAG	Normal
ON	ON	ON	OFF	OFF	ICUMX LPD	Normal
ON	ON	OFF	ON	ON	ICUMX JTAG	Core Sight
ON	ON	OFF	ON	OFF	ICUMX LPD	Core Sight
ON	OFF	ON	ON	ON	Core Sight	Normal
OFF	ON	ON	ON	ON	VDSP0	Normal
OFF	ON	ON	ON	OFF	VDSP1	Normal
OFF	ON	ON	OFF	ON	VDSP2	Normal
OFF	ON	ON	OFF	OFF	VDSP3	Normal
OFF	ON	OFF	ON	ON	Core Sight	VDSP0
OFF	ON	OFF	ON	OFF	Core Sight	VDSP1
OFF	ON	OFF	OFF	ON	Core Sight	VDSP2
OFF	ON	OFF	OFF	OFF	Core Sight	VDSP3
Other than above					Setting prohibited	

4.1.3. SW31 & SW33

SW31	1:MD14	2:MD13	3:MD29	4:MD32	5:MD31	6:X	7:MD16	8:X
DEFAULT	ON	ON	ON	ON	ON	ON	ON	ON
ON	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X	PullHigh	X
OFF	Open	Open	Open	Open	Open	X	Open	X
SW33	1:MD14	2:MD13	3:MD29	4:X	5:MD16	6:MD32	7:MD31	8:X
DEFAULT	ON	ON	ON	ON	OFF	ON	OFF	OFF
ON	GND	GND	GND	X	GND	GND	GND	X
OFF	Open	Open	Open	X	Open	Open	Open	X

◆ [SW33]

1:MD14	2:MD13	Selection of PLL Initial Multiplication Ratio
ON	ON	EXTAL input = 16.55 MHz, EXTAL Divider = x1/1
ON	OFF	EXTAL input = 20.00 MHz, EXTAL Divider = x1/1
OFF	ON	Setting prohibited
OFF	OFF	EXTAL input = 33.33 MHz, EXTAL Divider = x1/2

3:MD29	5:MD16	Selection mode
ON	-	Selects EXTALR as RCLK reference
OFF	-	Selects on chip oscillator (OCO) as RCLK reference
-	ON	Setting prohibited
-	OFF	Use this setting

6:MD32	7:MD31	Selects SCIF or HSCIF downloading mode
ON	ON	SCIF download
ON	OFF	HSCIF download 921600bps
OFF	ON	HSCIF download 1843200bps
Other than above		Setting prohibited

4.1.4. SW27 & SW29

SW27	1:MD27	2:MD22	3:X	4:X	5:X	6:X	7:X	8:X
------	--------	--------	-----	-----	-----	-----	-----	-----

DEFAULT	ON	ON	ON	ON	ON	ON	ON	ON
ON	PullHigh	PullHigh	X	X	X	X	X	X
OFF	Open	Open	X	X	X	X	X	X
SW29	1:MD27	2:MD22	3:X	4:X	5:X	6:X	7:X	8:X
DEFAULT	ON	ON	ON	ON	OFF	ON	ON	ON
ON	GND	GND	X	X	X	X	X	X
OFF	Open	Open	X	X	X	X	X	X

◆ [SW29]

1:MD27	2:MD22	Setting of pin multiplexing of DDR phy
ON	ON	LPDDR5
Other than above		Setting prohibited

4.1.5. SW30 & SW32

SW30	1:MD37	2:MD36	3:MD25	4:MD33	5:MD24	6:MD18	7:MD23	8:MD9
DEFAULT	ON	ON	ON	ON	ON	ON	ON	ON
ON	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh
OFF	Open	Open	Open	Open	Open	Open	Open	Open
SW32	1:MD37	2:MD36	3:MD25	4:MD23	5:MD33	6:MD24	7:MD18	8:MD9
DEFAULT	ON	ON	ON	ON	ON	ON	ON	ON
ON	GND	GND	GND	GND	GND	GND	GND	GND
OFF	Open	Open	Open	Open	Open	Open	Open	Open

◆ [SW32]

1:MD37	2:MD36	Selects PLL1 operation mode
ON	ON	Integer mode (SSCG OFF)
ON	OFF	Fractional mode (SSCG OFF)
OFF	ON	Reserved
OFF	OFF	Fractional mode, SSCG (Center spread $\pm 1.0\%$)

4:MD23	5:MD33	6:MD24	7:MD18	Selection mode
ON	-	-	-	Use this setting
OFF	-	-	-	Setting prohibited

-	ON	-	-	Use this setting
-	OFF	-	-	Setting prohibited
-	-	ON	-	Use this setting
-	-	OFF	-	Setting prohibited
-	-	-	ON	Use this setting
-	-	-	OFF	Setting prohibited

3:MD25	8:MD9	Selection mode
ON	-	Field BIST is not activated
OFF	-	Field BIST is activated
-	ON	The EXTAL pin is used for an oscillator
-	OFF	The EXTAL and XTAL pins are used for a resonator

4.2. PCIE MODE SWITCH

4.2.1. SW5

SW5	1:PORT CFG0	2:PORT CFG1	3:PORT CFG2	4:JTAG_ SEL_L	5:I2C_A DDR[0]	6:I2C_A DDR[1]	7:I2C_A DDR[2]	8:SMBU S_EN_L
DEFAULT	OFF	ON	OFF	OFF	OFF	OFF	OFF	ON
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:PORTCFG[0]	2:PORTCFG[1]	3:PORTCFG[2]	Function Mode
OFF	ON	ON	2Port - 16Lane
ON	OFF	ON	3Port - 16Lane
ON	OFF	ON	4Port - 16Lane
ON	ON	OFF	5Port - 16Lane
OFF	ON	OFF	8Port - 16Lane

5:I2C_ADDR[0]	6:I2C_ADDR[1]	7:I2C_ADDR[2]	I2C_ADDR = 0x1101[2:0]
ON	ON	ON	0x0: ADDR = 0x68
OFF	ON	ON	0x1: ADDR = 0x69
ON	OFF	ON	0x2: ADDR = 0x6A
OFF	OFF	ON	0x3: ADDR = 0x6B
ON	ON	OFF	0x4: ADDR = 0x6C
OFF	ON	OFF	0x5: ADDR = 0x6D
ON	OFF	OFF	0x6: ADDR = 0x6E
OFF	OFF	OFF	0x7: ADDR = 0x6F

4:JTAG_SEL_L	8:SMBUS_EN_L	Selection mode
ON	-	accessing PCIe PHY internal registers
OFF	-	controlling switch's boundary scan
-	ON	SMBUS protocol is selected.
-	OFF	I2C protocol is selected.

4.2.2. SW3

SW3	3:PCIE_MUX	2:PCIE_SEL	Connect
1-2	Don't Care	H(P3.3V)	A to C M.2 Key M(HSSSTP)
Center (DEFAULT)	Don't Care	L	A to B PCIe SW Connect
2-3	L	L	A to B PCIe SW Connect
	H	H	A to C M.2 Key M(HSSSTP)

4.3. SPI MODE SWITCH

4.3.1. SW34

SW34	1 : MSIOF2_SS2/MD25_SW	2 : GP1_25/SL_SW2_V
DEFAULT	OFF	OFF
ON	Pin LOW	Pin LOW
OFF	Pin HIGH(D3.3V_SPI)	Pin HIGH(D3.3V_SPI)

4.3.2. SW38

SW38	1 : AS_DOOR_SW	2 : GP1_25/SL_SW2_V
DEFAULT	OFF	OFF
ON	Pin LOW	Pin LOW
OFF	Pin HIGH(VIN_5V)	Pin HIGH(VIN_5V)

4.4. ETHERNET MODE SWITCH

4.4.1. SW18 (88Q6113)

SW18	1:RMU_SEL[0]	2:RMU_SEL[1]	3:RMU_SEL[2]	4:P9_SMODE[0]	5:P9_SMODE[1]	6:P10_SMODE[0]	7:P10_SMODE[1]	8:X
DEFAULT	ON	ON	ON	ON	ON	ON	ON	ON
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	X
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X

1:RMU_SEL[0]	2:RMU_SEL[1]	3:RMU_SEL[2]	Selection RMU_SEL[2:0]
ON	ON	ON	0x0: PORT 1 (SGMII)
OFF	ON	ON	0x1: PORT 7 (xMII)
ON	OFF	ON	0x2: PORT 8 (xMII)
OFF	OFF	ON	0x3: PORT 9 (XFI)
OFF	OFF	OFF	0x7: RMU disable

4:P9_SMODE[0]	5:P9_SMODE[1]	Selection P9_SMODE[1:0]
ON	ON	0x0: 1000BASE-X/SGMII
OFF	ON	0x1: 5GBASE-R
ON	OFF	0x2: USXGMII
OFF	OFF	0x3: 10GBASE-R

6:P10_SMODE[0]	7:P10_SMODE[1]	Selection P10_SMODE[1:0]
ON	ON	0x0: 1000BASE-X/SGMII
OFF	ON	0x1: 5GBASE-R
ON	OFF	0x2: USXGMII
OFF	OFF	0x3: 10GBASE-R

4.4.2. SW19 (88Q6113)

SW19	1:P7 MODE0	2:P8 MODE0	3:P1 SMODE	4:P2 SMODE	5:P3 SMODE	6:C3 LED_RES V (P11_0)	7:P8 OUTEN_RE SV (P11_1)	8:CPU_ EN
DEFAULT	ON	OFF	OFF	OFF	OFF	ON	ON	OFF
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:P7 MODE0	2:P8 MODE0	3:P1 SMODE	4:P2 SMODE	5:P3 SMODE	Selection mode
ON	-	-	-	-	PORT 7 dsiable
OFF	-	-	-	-	PORT 7 RGMII
-	ON	-	-	-	PORT 8 dsiable
-	OFF	-	-	-	PORT 8 RGMII
-	-	ON	-	-	100FX
-	-	OFF	-	-	1000BASE-X/SGMII
-	-	-	ON	-	100FX
-	-	-	OFF	-	1000BASE-X/SGMII
-	-	-	-	ON	100FX
-	-	-	-	OFF	1000BASE-X/SGMII

6:C3 LED_RESV (P11_0)	7:P8 OUTEN_RESV (P11_1)	Selection P11_SMODE[1:0]
ON	ON	0x0: P11 PCIe
OFF	ON	0x1: P11 PCIe
ON	OFF	0x2: P11 Eth port in 1000BASE-X/SGMII
OFF	OFF	0x3: P11 Eth port with 2500BASE-X

8:CPU_EN	Selection mode
ON	CPU Disable
OFF	CPU Enable

4.4.3. SW20 (88Q6113)

SW20	1:ADDR0	2:ADDR1	3:ADDR2	4:ADDR3	5:ACTIVE0	6:ACTIVE1	7:X	8:X
DEFAULT	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
ON	LOW	LOW	LOW	LOW	LOW	LOW	X	X
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X	X

1:ADDR0	2:ADDR1	3:ADDR2	4:ADDR3	Selection ADDR[3:0]
ON	ON	ON	ON	MULTI-CHIP SMI ADDR 0xF
OFF	ON	ON	ON	MULTI-CHIP SMI ADDR 0xE
ON	OFF	ON	ON	MULTI-CHIP SMI ADDR 0xD
OFF	OFF	ON	ON	MULTI-CHIP SMI ADDR 0xC
ON	ON	OFF	ON	MULTI-CHIP SMI ADDR 0xB
OFF	ON	OFF	ON	MULTI-CHIP SMI ADDR 0xA
ON	OFF	OFF	ON	MULTI-CHIP SMI ADDR 0x9
OFF	OFF	OFF	ON	MULTI-CHIP SMI ADDR 0x8
ON	ON	ON	OFF	MULTI-CHIP SMI ADDR 0x7
OFF	ON	ON	OFF	MULTI-CHIP SMI ADDR 0x6
ON	OFF	ON	OFF	MULTI-CHIP SMI ADDR 0x5
OFF	OFF	ON	OFF	MULTI-CHIP SMI ADDR 0x4
ON	ON	OFF	OFF	MULTI-CHIP SMI ADDR 0x3
OFF	ON	OFF	OFF	MULTI-CHIP SMI ADDR 0x2
ON	OFF	OFF	OFF	MULTI-CHIP SMI ADDR 0x1
OFF	OFF	OFF	OFF	SINGLE CHIP SMI ADDR 0x0

5:ACTIVE0	6:ACTIVE1	Selection mode
ON	-	PORT 1 – 6 disable
OFF	-	PORT 1 – 6 enable
-	ON	PORT 7 – 10 disable
-	OFF	PORT 7 – 10 enable

4.4.4. SW21 (88Q6113)

SW21	1 : Q6113_2.5V_VDDO	2 : GND	Selection mode
	ON	ON	Setting prohibited
	ON	OFF	BSCAN mode
DEFAULT	OFF	ON	DAP mode
	OFF	OFF	Floating

4.4.5. SW22 (88Q5040)

SW22	1:P5_M ODE[0]	2:P5_M ODE[1]	3:P5_M ODE[2]	4:P5_V DDOS	5:P6_M ODE[0]	6:P6_M ODE[1]	7:P6_M ODE[2]	8:P6_V DDOS
DEFAULT	OFF	OFF	OFF	ON	OFF	OFF	ON	ON
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:P5_MODE[0]	2:P5_MODE[1]	3:P5_MODE[2]	P5_MODE[2:0]
ON	ON	ON	0x0: MII MAC MODE
OFF	ON	ON	0x1: MII PHY
ON	OFF	ON	0x2: MII MAC
OFF	OFF	ON	0x3: RESV
ON	ON	OFF	0x4: RMII PHY
OFF	ON	OFF	0x5: RMII MAC
ON	OFF	OFF	0x6: DISABLE
OFF	OFF	OFF	0x7: RGMII

4:P5_VDDOS	Selection P5_VDDO
ON	P5_VDDO = 2.5V
OFF	P5_VDDO = 3.3V

5:P6_MODE[0]	6:P6_MODE[1]	7:P6_MODE[2]	P6_MODE[2:0]
ON	ON	ON	0x0: MII MAC MODE
OFF	ON	ON	0x1: MII PHY
ON	OFF	ON	0x2: MII MAC
OFF	OFF	ON	0x3: RESV
ON	ON	OFF	0x4: RMII PHY
OFF	ON	OFF	0x5: RMII MAC
ON	OFF	OFF	0x6: DISABLE
OFF	OFF	OFF	0x7: RGMII

8:P6_VDDOS	Selection P6_VDDO
ON	P6_VDDO = 2.5V
OFF	P6_VDDO = 3.3V

4.4.6. SW23 (88Q5040)

SW23	1:ADDR	2:ADDR	3:ADDR	4:ADDR	5:ACTIV	6:GPIO_	7:SPI_V	8:CPU_
------	--------	--------	--------	--------	---------	---------	---------	--------

	[0]n	[1]n	[2]n	[3]n	E	VDDOS	DDOS	EN
DEFAULT	OFF	OFF	OFF	OFF	OFF	ON	ON	OFF
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:ADDR[0]n	2:ADDR[1]n	3:ADDR[2]n	4:ADDR[3]n	Selection ADDR[3:0]
ON	ON	ON	ON	MULTI-CHIP SMI ADDR 0xF
OFF	ON	ON	ON	MULTI-CHIP SMI ADDR 0xE
ON	OFF	ON	ON	MULTI-CHIP SMI ADDR 0xD
OFF	OFF	ON	ON	MULTI-CHIP SMI ADDR 0xC
ON	ON	OFF	ON	MULTI-CHIP SMI ADDR 0xB
OFF	ON	OFF	ON	MULTI-CHIP SMI ADDR 0xA
ON	OFF	OFF	ON	MULTI-CHIP SMI ADDR 0x9
OFF	OFF	OFF	ON	MULTI-CHIP SMI ADDR 0x8
ON	ON	ON	OFF	MULTI-CHIP SMI ADDR 0x7
OFF	ON	ON	OFF	MULTI-CHIP SMI ADDR 0x6
ON	OFF	ON	OFF	MULTI-CHIP SMI ADDR 0x5
OFF	OFF	ON	OFF	MULTI-CHIP SMI ADDR 0x4
ON	ON	OFF	OFF	MULTI-CHIP SMI ADDR 0x3
OFF	ON	OFF	OFF	MULTI-CHIP SMI ADDR 0x2
ON	OFF	OFF	OFF	MULTI-CHIP SMI ADDR 0x1
OFF	OFF	OFF	OFF	SINGLE CHIP SMI ADDR 0x0

5:ACTIVE	6:GPIO_VDDOS	7:SPI_VDDOS	8:CPU_EN	Selection mode
ON	-	-	-	PORT 1 – 6 Disable
OFF	-	-	-	PORT 1 – 6 Forward
-	ON	-	-	GPIO_VDDO = 2.5V
-	OFF	-	-	GPIO_VDDO = 3.3V
-	-	ON	-	SPI_VDDO = 2.5V
-	-	OFF	-	SPI_VDDO = 3.3V
-	-	-	ON	CPU Disable
-	-	-	OFF	CPU Enable

4.4.7. SW24 (88Q5040)

SW24	1:RMU_	2:RMU_	3:RMU_	4:P1_M	5:P2_M	6:P3_M	7:P4_M	8:X
------	--------	--------	--------	--------	--------	--------	--------	-----

	SEL[0]	SEL[1]	SEL[2]	ASTER	ASTER	ASTER	ASTER	
DEFAULT	ON	ON	ON	OFF	OFF	OFF	OFF	OFF
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	X
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	X

1:RMU_SEL[0]	2:RMU_SEL[1]	3:RMU_SEL[2]	RMU_SEL [2:0]
ON	ON	ON	0x0: PORT 1 (100BT1)
OFF	ON	ON	0x1: PORT 4 (100BT1/BTX/SGMII)
ON	OFF	ON	0x2: PORT 5 (xMII)
OFF	OFF	ON	0x3: RESV
ON	ON	OFF	0x4: RESV
OFF	ON	OFF	0x5: RESV
ON	OFF	OFF	0x6: RESV
OFF	OFF	OFF	0x7: RMU Disable

4:P1_MASTER	5:P2_MASTER	6:P3_MASTER	7:P4_MASTER	Selection mode
ON	-	-	-	SLAVE
OFF	-	-	-	MASTER
-	ON	-	-	SLAVE
-	OFF	-	-	MASTER
-	-	ON	-	SLAVE
-	-	OFF	-	MASTER
-	-	-	ON	SLAVE
-	-	-	OFF	MASTER

4.4.8. SW25 (88Q5040)

SW25	1:MODE [0]	2:MODE [1]	3:MODE [2]	4:JTAG_ EN	5:SSPI_E N	6:SLEEP_ CAP_DIS	7:SSPI_M ODE[0]	8:SSPI_M ODE[1]
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DEFAULT	OFF	OFF	OFF	ON	OFF	OFF	OFF	ON
ON	LOW	LOW	LOW	LOW	LOW	LOW	LOW	LOW
OFF	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh	PullHigh

1:MODE[0]	2:MODE[1]	3:MODE[2]	PORT 4 MODE[2:0]	PORT 5 MODE[2:0]
ON	ON	ON	0x0: SGMII/1000X	0x0: RGMII
OFF	ON	ON	0x1: 100BASE-TX	0x1: 100BASE-FX
ON	OFF	ON	0x2: 100BASE-TX	0x2: SGMII/1000X
OFF	OFF	ON	0x3: 100BASE-TX	0x3: RGMII
ON	ON	OFF	0x4: 100BASE-FX	0x4: RGMII
OFF	ON	OFF	0x5: 100BASE-T1	0x5: 100BASE-FX
ON	OFF	OFF	0x6: 100BASE-T1	0x6: SGMII/1000X
OFF	OFF	OFF	0x7: 100BASE-T1	0x7: RGMII

4:JTAG_EN	5:SSPI_EN	6:SLEEP_CAP_DIS	Selection mode
ON	-	-	NORMAL OPERATION
OFF	-	-	JTAG ENABLE
-	ON	-	Slave SPI
-	OFF	-	SMI_CPU
-	-	ON	TC10 Enable
-	-	OFF	TC10 Disable

7:SSPI_MODE[0]	8:SSPI_MODE[1]	Selection SSPI_MODE[1:0]
ON	ON	0x0: Clock Polarity = 0, Clock Phase = 0
OFF	ON	0x1: Clock Polarity = 0, Clock Phase = 1
ON	OFF	0x2: Clock Polarity = 1, Clock Phase = 0
OFF	OFF	0x3: Clock Polarity = 1, Clock Phase = 1

4.5. DISPLAY MODE SWITCH

4.5.1. SW2 (DSI0)

SW2	3:SEL_DSI0	2:SELDSI0_R	Connect
1-2	Don't Care	H(D3.3V1_DSI)	A to C MAX96789 Connect
Center(DEFAULT)	Don't Care	L	A to B SN65DSI86 Connect
2-3	L	L	A to B SN65DSI86 Connect
	H	H	A to C MAX96789 Connect

4.5.2. SW44 (DSI1)

SW44	3:SEL_DSI1	2:SELDSI1_R	Connect
1-2	Don't Care	H(D3.3V1_DSI)	A to C MAX96789 Connect
Center(DEFAULT)	Don't Care	L	A to B LT9611 Connect
2-3	L	L	A to B LT9611 Connect
	H	H	A to C MAX96789 Connect

4.6. CAN MODE SWITCH

4.6.1. SW(7、8、9、10、13、14)

SW7	SW8	SW9	SW10	SW13	SW14	Selection mode
PIN 2 – 1	PIN 2 – 3	PIN 2 – 1	PIN 2 – 1	PIN 2 – 1	PIN 2 – 1	CANFD1 CANFD5
PIN 2 – 3	PIN 2 – 1	PIN 2 – 3	PIN 2 – 3	PIN 2 – 3	PIN 2 – 3	FlexRay A FlexRay B

4.6.2. SW11

SW11 : CANFD5	1 : CAN_SILENT
DEFAULT	ON
ON	Short
OFF	Open

4.6.3. SW12

SW12	1 : RXD	2 : FXR_TXENB_D	Selection SW12
	ON	ON	Setting prohibited
DEFAULT	ON	OFF	CANFD5 RXD
	OFF	ON	FlexRay B TXEN
	OFF	OFF	Floating

4.6.4. SW15

SW15	1 : TXD	2 : FXR_TXENA_D	Selection SW15
	ON	ON	Setting prohibited
DEFAULT	ON	OFF	CANFD5 TXD
	OFF	ON	FlexRay A TXEN
	OFF	OFF	Floating

4.7. FLEXRAY MODE SWITCH

4.7.1. SW16

SW16 : FlexRay Transceiver Channel A	1 : EN
DEFAULT	OFF
ON	Short (CAN_D3.3V_FLEX)
OFF	Open

4.7.2. SW17

SW17 : FlexRay Transceiver Channel B	1 : EN
DEFAULT	OFF
ON	Short (CAN_D3.3V_FLEX)
OFF	Open

4.8. MCU MODE SWITCH

4.8.1. SW39

SW39	1 : ACC_SW
DEFAULT	OFF
ON	ACC_SW_MCU HIGH
OFF	ACC_SW_MCU LOW

4.8.2. SW42

SW42	1 : MCU_MXBKUPN
DEFAULT	OFF
ON	MxBKUP = 1
OFF	MxBKUP = 0

4.8.3. SW43

SW43	1 : TJA1463AT WAKE pin
DEFAULT	OFF
ON	WAKE OFF
OFF	WAKE ON

5. BUTTON

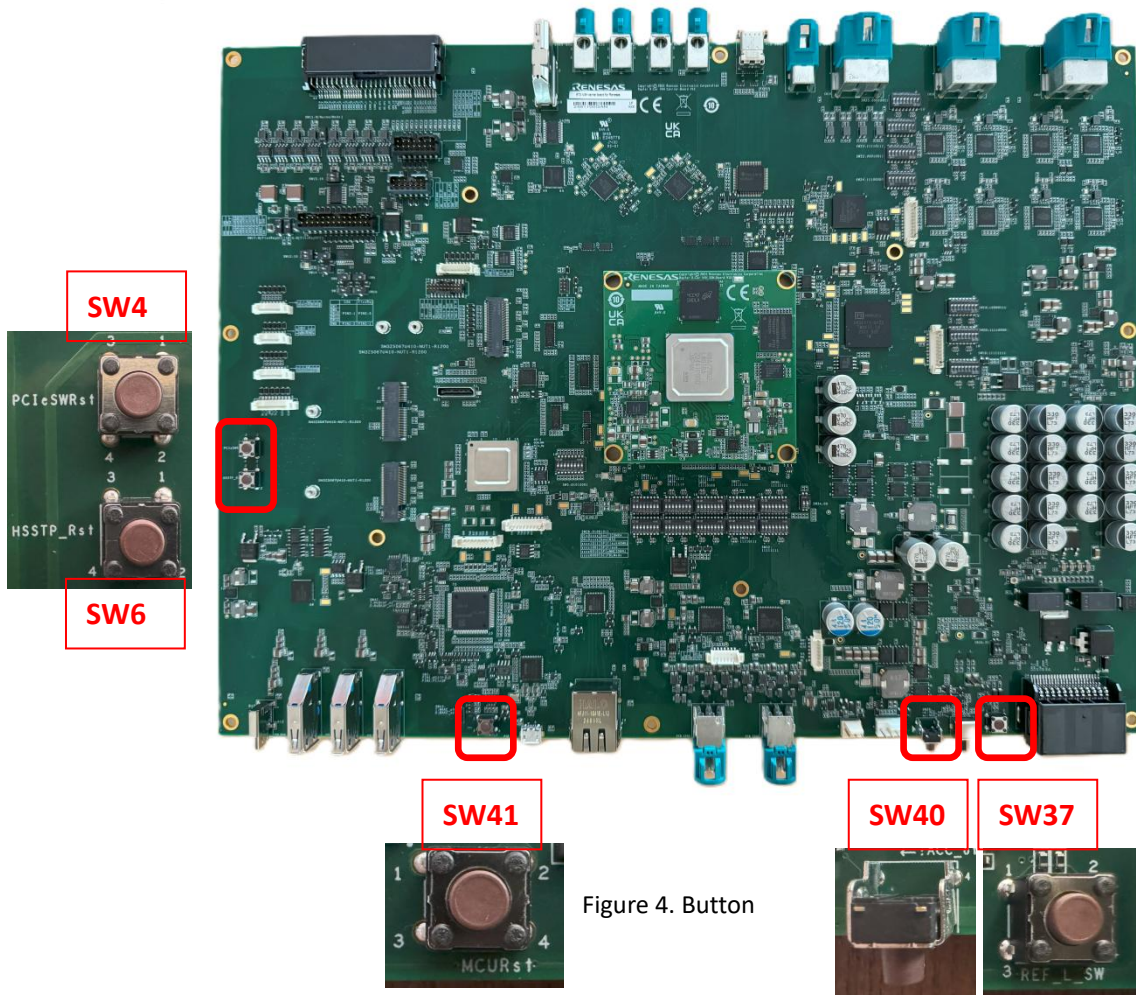


Figure 4. Button

SW40	Reset Switch
SW37	Reference line sw
SW41	MCU Reset
SW4	PCIe SW Reset
SW6	PCIe CN Reset

6. Variable Voltage

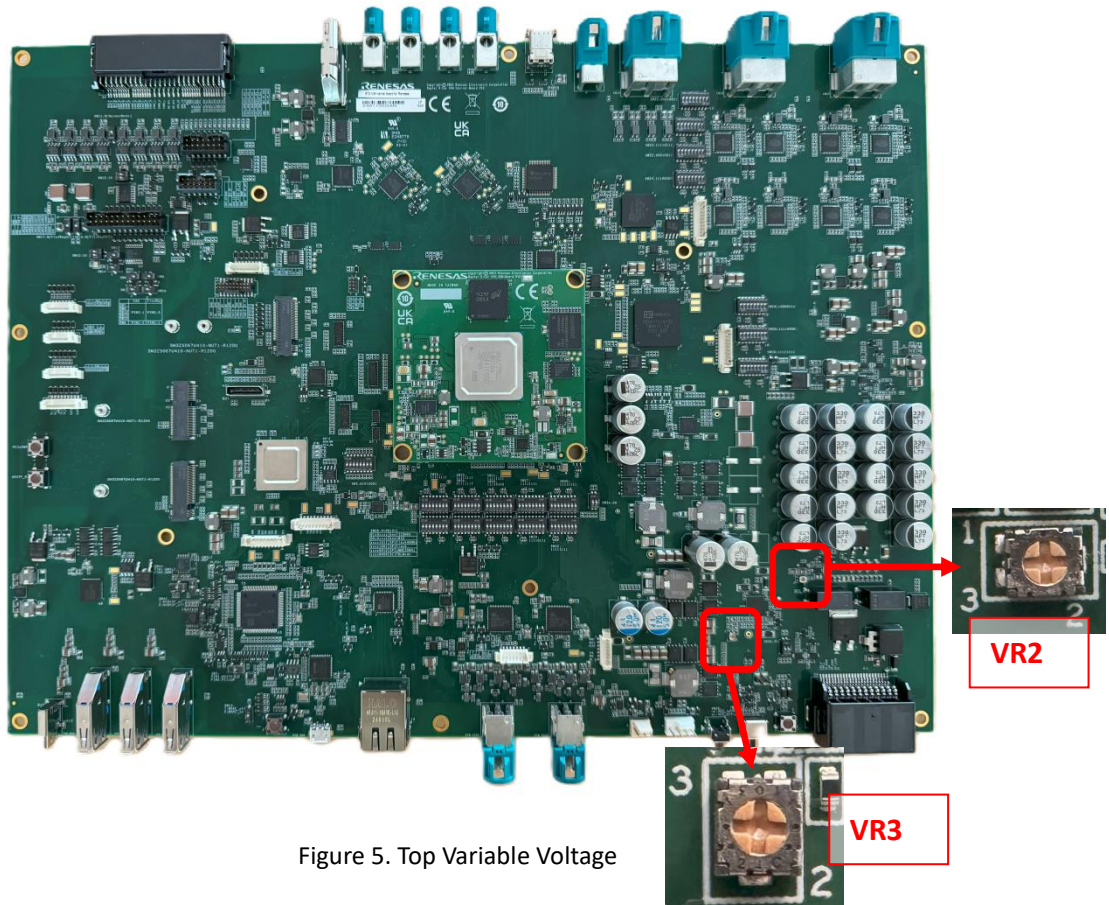


Figure 5. Top Variable Voltage

VR2	VR3
VIN_AFTER_DIODE IG1_CN R1600 4.7K 1% 1005 VR2 10k 20% 0.1W	PM1B_FB2 R1688 68.1K 1% 1005 C1582 10pF 1005 50V R1691 NI_7.5K 1% 1005 R1692 4.7K 1% 1005 VR3 10k 20% 0.1W ot Stuff AGND2 AGND2 R1692(4.7K), VR3 Mount/R1691(NI_7.5K) Dismount(default) Variable Voltage R1692(4.7K), VR3 Dismount/R1691(NI_7.5K) Mount Fixed Voltage 8.064V $V_{OUT} = (68.1k/7.5k + 1) \times 0.8 = 8.06$ 4V

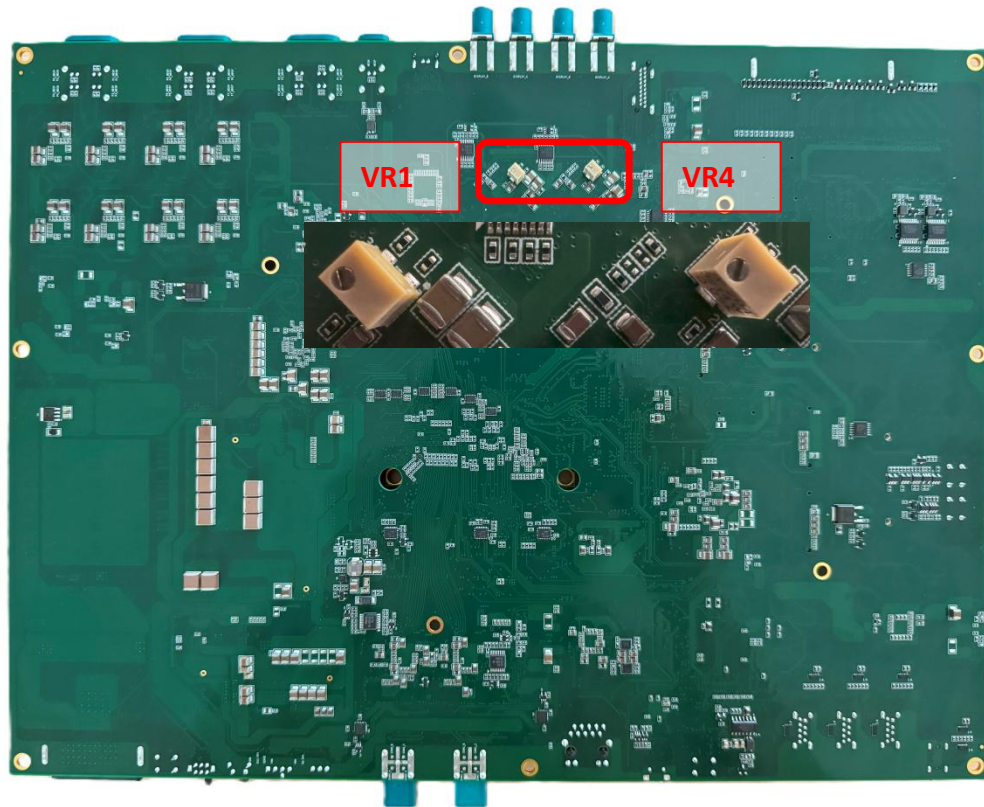


Figure 6. Bottom Variable Voltage

VR1 : U28 MAX96789GTN/V+

Color mode, RCLKOUT settings

[Design Note] CFG2(CP9) voltage - setting table (@percentage of SPI_D3.3V)

MIN(%)	TYP(%)	MAX(%)	Color mode	RCLKOUT
0.0	0.0	11.7		RCLKOUT disabled
16.9	20.2	23.6	24-bit mode	RCLKOUT enabled
28.8	32.1	35.5		RCLKOUT disabled
40.7	44.0	47.4	32-bit mode	RCLKOUT enabled
52.6	56.0	59.3	27-bit mode	RCLKOUT disabled
64.5	67.9	71.2	(High BW mode)	RCLKOUT enabled
76.4	79.8	83.1		
88.3	100	100		Invalid

VR4 : U181 MAX96789GTN/V+

Color mode, RCLKOUT settings

[Design Note] CFG2(CP9) voltage - setting table (@percentage of SPI_D3.3V)

MIN(%)	TYP(%)	MAX(%)	Color mode	RCLKOUT
0.0	0.0	11.7		RCLKOUT disabled
16.9	20.2	23.6	24-bit mode	RCLKOUT enabled
28.8	32.1	35.5		RCLKOUT disabled
40.7	44.0	47.4	32-bit mode	RCLKOUT enabled
52.6	56.0	59.3	27-bit mode	RCLKOUT disabled
64.5	67.9	71.2	(High BW mode)	RCLKOUT enabled
76.4	79.8	83.1		
88.3	100	100		Invalid

7. APPENDIX